

Model predictive control of vehicle dynamics based on the Koopman operator with extended dynamic mode decomposition

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Abstract—The control of vehicle dynamics is a very demanding task due to the complex nonlinear tire characteristics and the coupled lateral and longitudinal dynamics of the vehicle. When designing a Model Predictive Controller (MPC) for vehicle dynamics, this can lead to a non-convex optimization problem. A novel approach to solve the problem of controlling nonlinear systems is based on the so-called Koopman operator. The Koopman operator is a linear operator that governs the evolution of scalar functions (often referred to as observables) along the trajectories of a given nonlinear dynamical system and is a powerful tool for the analysis and decomposition of nonlinear dynamical systems. The main idea is to lift the nonlinear dynamics to a higher dimensional space where its evolution can be described with a linear system model. In this paper we propose a model predictive controller for vehicle dynamics based on the Koopman operator decomposition of vehicle dynamics with Extended Dynamic Mode Decomposition (EDMD) method. Both model identification and predictive controller design are validated using Matlab/Simulink environment.

Index Terms—Koopman operator, basis function, data-driven methods, extended dynamic mode decomposition, model predictive control, vehicle dynamics

I. INTRODUCTION

As the name suggests, Model Predictive Control (MPC) is based on a model of a system for predicting the future trajectory and determining the optimal input sequence. Apart from the dynamical model of a system, it also takes into account its constraints and it is the only advanced control strategy that has a significant and far-reaching impact on the industry [1].

Like most of the systems surrounding us, a vehicle is a nonlinear dynamical system with complicated tire characteristics and coupled lateral and longitudinal dynamics. For this reason, it can be quite challenging to build a model that is able to accurately describe the behavior of a vehicle in all its working regions. In [2] the authors present and compare the three most common modeling approaches: 1. nonlinear physical modeling, 2. linear physical modeling and 3. data-driven model. The approaches 1. and 2. are based on Newton's second law and the empirical understanding of certain driving conditions, making them easy to understand. On the other hand, 3. relies on simulated and/or measured data from which

a general parametric model is derived, without any previously imposed structure. The results show that under normal driving conditions, data-driven models are better able to predict the behavior of the vehicle than physically derived models.

When used in MPC, nonlinear models lead to non-convex optimization problems with multiple local optima, which requires much more computational power and offers no guarantees for global optimality. Therefore the use of linear representations is preferred. Here we focus on the Koopman operator, a linear operator that governs the evolution of scalar functions (often referred to as observables) along the trajectories of a given nonlinear dynamical system [3]. In recent years, increasing efforts have been made to study the Koopman operator methods for state estimation [4] and control. More details about the optimal control based on the Koopman operator can be found in [5], while detailed theoretical explanations of MPC using a linear model derived with the Koopman operator (KMPC) are reported in [6]. So far, KMPC has been used in various engineering applications, such as continuously stirred tank reactor process control [7], power grid transient stabilization [8], nonlinear flow control [9] and trajectory tracking of soft robots [10].

The Koopman operator was used for the data-driven identification of vehicle dynamics in [11], where two different approaches for the selection of basis functions were compared: the Extended Dynamic Mode Decomposition (EDMD) and the eigenfunction approach. Both approaches are presented with a numerical example and their feasibility is discussed. However, the derived model was not used for controller synthesis. In [12] the authors went one step further and developed a framework for vehicle dynamics identification using the Koopman operator with deep learning methods. The proposed modeling method was tested by using a linear MPC with steering angle and engine throttle as inputs. The results showed a satisfactory tracking performance, which proves the effectiveness and robustness of the proposed approach.

In this paper we apply the EDMD identification approach presented in [11] to the single-track vehicle model with front wheel steering and extend it further with the linear MPC design. Unlike in [12], we use tire slip on both wheels as control inputs in addition to the steering angle. The rest of

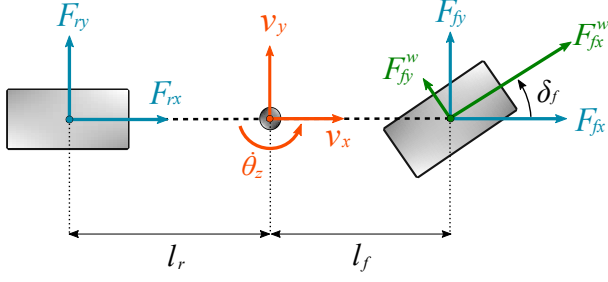


Fig. 1. The single-track vehicle model.

this paper is structured as follows. In Section II, a single-track vehicle model is presented. Section III gives a brief theoretical overview of the Koopman operator and describes vehicle identification which was performed to obtain the model approximation. In Section IV MPC design is described and simulation results are presented. Finally, some conclusions are drawn in Section V.

II. VEHICLE MODEL

Here we present the single-track model based on the models described in [11] and [13]. The mark $\bullet \in \{f, r\}$ represents the values for front and rear wheel. The model is shown in Fig. 1 and is given by the following equations:

$$m\dot{v}_x = m\omega_z v_y + F_{fx} + F_{rx} - \frac{1}{2}c_w\rho A_w v_x \sqrt{v_x^2 + v_y^2}, \quad (1)$$

$$m\dot{v}_y = -m\omega_z v_x + F_{fy} + F_{ry} - \frac{1}{2}c_w\rho A_w v_y \sqrt{v_x^2 + v_y^2}, \quad (2)$$

$$J_z\ddot{\theta}_z = F_{fy}l_f - F_{ry}l_r, \quad (3)$$

$$J_w\dot{\omega}_\bullet = T_\bullet - r_w F_{\bullet x}^w. \quad (4)$$

In (1) - (4) v_x stands for longitudinal velocity, v_y for lateral velocity, $\dot{\theta}_z$ for yaw rate and $\omega_\bullet$ for wheel angular velocity. Mechanical parameters are mass m , moment of inertia around the yaw axis J_z , longitudinal distance of the front axle from the Centre of Mass (CM) l_f , longitudinal distance of the rear axle from the CM l_r , wheel moment of inertia J_w and wheel radius r_w . Air-resistance parameters are drag coefficient c_w , air density ρ and the total area exposed to the air flow A_w . The forces along the longitudinal and lateral axes of the vehicle are calculated using the tire forces as

$$\begin{aligned} F_{fx} &= F_{fx}^w \cos \delta_f - F_{fy}^w \sin \delta_f, & F_{rx} &= F_{rx}^w, \\ F_{fy} &= F_{fx}^w \sin \delta_f + F_{fy}^w \cos \delta_f, & F_{ry} &= F_{ry}^w. \end{aligned} \quad (5)$$

The steering angle is marked with δ_f . Longitudinal and lateral tire forces $F_{fx}^w, F_{fy}^w, F_{rx}$ and F_{ry} depend on the slip angles α_f and α_r , longitudinal slips s_{fx} and s_{rx} , friction coefficients μ_f and μ_r and vertical load forces F_{fz} and F_{rz} . Tire forces are given by a nonlinear model such as Magic Formula [14] or TMeasy [15] and can be written as

$$\begin{aligned} F_{\bullet x}^w &= f_x(\alpha_\bullet, s_{\bullet x}, \mu_\bullet, F_{\bullet z}), \\ F_{\bullet y}^w &= f_y(\alpha_\bullet, s_{\bullet x}, \mu_\bullet, F_{\bullet z}), \end{aligned} \quad (6)$$

with tire slips

$$s_{\bullet x} = \frac{r_w \omega_\bullet}{v_{\bullet x}^w} - 1, \quad v_{\bullet x}^w \neq 0 \quad (7)$$

and tire slip angles

$$\alpha_\bullet = \arctan \left(\frac{v_{\bullet y}^w}{v_{\bullet x}^w} \right). \quad (8)$$

Wheel velocities are

$$\begin{aligned} v_{fx}^w &= v_{fx} \cos \delta_f + v_{fy} \sin \delta_f, & v_{rx}^w &= v_{rx}, \\ v_{fy}^w &= -v_{fx} \sin \delta_f + v_{fy} \cos \delta_f, & v_{ry}^w &= v_{ry}, \end{aligned} \quad (9)$$

where

$$\begin{aligned} v_{fx} &= v_x, & v_{fy} &= v_y + l_f \dot{\theta}_z, \\ v_{rx} &= v_x, & v_{ry} &= v_y - l_r \dot{\theta}_z. \end{aligned} \quad (10)$$

The values of the vertical load forces $F_{\bullet z}$ in (6) depend on the dynamics of the load transfer. The model (1)-(10) can be extended with effects such as roll and pitch dynamics, i.e. additional degrees of freedom. Extended models can be found e.g. in [16], [17]. Here we omit these extensions because they would complicate the calculations but would not contribute to the algorithms we are testing.

A. Three state single track model with linear tires

One of the possible simplifications of the model is to omit (4) and (7), which then reduces model complexity and leads to a model with only three states: $\mathbf{x} = [v_x \ v_y \ \dot{\theta}_z]^T$. Additionally, nonlinear tire forces (6) are represented by a linear model

$$F_{\bullet x}^w = C_x s_{\bullet x}, \quad F_{\bullet y}^w = -C_y \alpha_\bullet, \quad (11)$$

where C_x and C_y stand for the longitudinal and cornering stiffness of the tire, respectively. The input vector becomes $\mathbf{u} = [\delta_f \ s_{fx} \ s_{rx}]^T$ and the existence of a low-level torque controller that controls the current slip at each wheel to the desired slip value is assumed, reducing the computational complexity of the model [18]. As mentioned in [16], this tire model is valid for small slip and slip angle values that can be achieved using slip and slip angle controllers. This model is used to convert input forces to steering angles and tire slips in IV-C. The model parameters are specified in TABLE I.

B. Three state single track model without tire model

If the model of a system is approximated using the EDMD, state derivatives and inputs must have a linear relationship due to the structure of the method [6]. For this to be the case, (5) - (7) are neglected and the input vector is replaced by $\mathbf{u} = [F_{fx} \ F_{fy} \ F_{rx}]^T$. This model is used for system identification in III-C and MPC system design in IV-A.

TABLE I
VEHICLE PARAMETERS

Parameter	Description	Value	Unit
m	mass of the vehicle	1752	kg
l_f	front axle to CM distance	1.435	m
l_r	rear axle to CM distance	1.31	m
J_z	moment of inertia around yaw axis	2286	kg · m ²
c_w	drag coefficient	0.31	-
ρ	air density	1.2	kg/m ³
A_w	surface exposed to the air flow	2.2	m ²
C_x	longitudinal tire stiffness	87712	N
C_y	lateral tire stiffness	51488	N/rad

III. KOOPMAN OPERATOR

The Koopman operator is a (typically infinite-dimensional) linear predictor for non-linear dynamical systems. The basic idea is to lift (or embed) the nonlinear dynamics into a higher dimensional space where its evolution is approximately linear [6].

A. Koopman operator basics

Consider an uncontrolled discrete-time dynamical system

$$\begin{aligned} \mathbf{x}_{t+1} &= \mathbf{f}(\mathbf{x}_t) \\ \mathbf{y}_t &= \mathbf{g}(\mathbf{x}_t) \end{aligned} \quad (12)$$

where $\mathbf{x}_t$ is the current state and $\mathbf{x}_{t+1}$ is the next state. Then the Koopman operator $\mathcal{K}$ acts on observable functions ψ as

$$\mathcal{K}\psi(\mathbf{x}_t) = \psi(\mathbf{f}(\mathbf{x}_t)) = \psi(\mathbf{x}_{t+1}). \quad (13)$$

The Koopman operator $\mathcal{K}$ describes the nonlinear dynamical system (12) completely, provided that the corresponding higher dimensional state space contains the components of the original state space.

For control purposes, it is essential to include input signals into (12) - (13). One way to do this is to extend the state space with the space of all control sequences $\chi_t = [\mathbf{x}_t \ \mathbf{u}_t]^T$. Given the dynamics of the controlled system

$$\begin{aligned} \mathbf{x}_{t+1} &= \mathbf{f}(\mathbf{x}_t, \mathbf{u}_t) \\ \mathbf{y}_t &= \mathbf{g}(\mathbf{x}_t) \end{aligned} \quad (14)$$

the augmented state evolution is

$$\chi_{t+1} = \mathbf{h}(\chi_t) = \begin{bmatrix} \mathbf{f}(\mathbf{x}_t, \mathbf{u}_0) \\ \mathcal{S}\mathbf{u} \end{bmatrix}, \quad (15)$$

where $\mathcal{S}$ is the left shift operator, i.e. $\mathcal{S}\mathbf{u}_t = \mathbf{u}_{t+1}$.

Since the finite-dimensional representation of the Koopman operator can rarely be determined, data-driven strategies, such as the EDMD [6] or the eigenfunctions [19] are used.

B. Extended Dynamic Mode Decomposition

In this paper we use the EDMD to construct a finite-dimensional approximation of the Koopman operator $\mathcal{K}$. For the uncontrolled system (12), the approximation is done by solving the optimization problem

$$\min_A \sum_{j=1}^K \|\Psi(\mathbf{x}_{t+1}^j) - A\Psi(\mathbf{x}_t^j)\|_2^2, \quad (16)$$

where $\mathbf{z}_t^j = \Psi(\mathbf{x}_t^j) = [\psi_1(\mathbf{x}_t^j) \ \dots \ \psi_{n_\psi}(\mathbf{x}_t^j)]^T$ is a vector of lifting functions (observables) for the j th sample. The states $\mathbf{x}_t$ and $\mathbf{x}_{t+1}$ are obtained by simulating the nonlinear system model and K is the cardinality of the simulated dataset (number of simulated time steps). Important to note is that the states may, but do not have to be a part of a trajectory. It is sufficient to perform K different one-step simulations with random samples from the state space, i.e. from the intervals of interest which are subsets of the state space, as initial conditions.

For the controlled system (14), the optimization problem (16) becomes

$$\min_{A,B} \sum_{j=1}^K \|\Psi(\mathbf{x}_{t+1}^j) - (A\Psi(\mathbf{x}_t^j) - B\mathbf{u}_t^j)\|_2^2. \quad (17)$$

The matrices A and B define a discrete linear system approximation of the dynamics (14):

$$\begin{aligned} \mathbf{z}_{t+1} &= A\mathbf{z}_t + B\mathbf{u}_t \\ \hat{\mathbf{y}}_t &= C\mathbf{z}_t, \end{aligned} \quad (18)$$

where $\hat{\mathbf{y}}_t$ is the estimate of the output. The matrix C can be obtained by minimizing the least square cost

$$\min_C \sum_{j=1}^K \|\mathbf{y}_t^j - C\Psi(\mathbf{x}_t^j)\|_2^2. \quad (19)$$

From (18) it can be seen that the approximation requires that the observables depend linearly on the input, as already emphasized in II-B.

C. Vehicle dynamics identification

To identify a vehicle dynamics model, we first have to select the learning dataset. All the states were uniformly sampled from predetermined intervals: $v_x \in [10 \ 50]m/s$, $v_y \in [-30 \ 30]m/s$ and $\dot{\theta}_z \in [-10 \ 10]rad/s$, while the forces were uniformly sampled in the interval $[-F_m \ F_m]$ with $F_m \in \{5, 50, 500, 5000\}N$. The number of samples for each interval is $N_s = 20$. When combined into all possible combinations, the columns of the sampled vectors result in $4 \cdot 20^4$ values in the dataset.

The next step is to select the basis functions $\psi(\cdot)$, i.e. the observables. Since there is no general recommendation for the selection of the basis function vector, polynomial basis functions were chosen. A polynomial basis consists of monomials of the state vector elements:

$$B_k = \{v_x^a \cdot v_y^b \cdot \dot{\theta}_z^c | a, b, c \in \{0, 1, 2, \dots, k\}\}, \quad (20)$$

where k is the order of the basis B_k . Simulations with bases of different orders were performed and the averaged root mean square error (RMSE) was calculated:

$$RMSE = 100 \frac{\sqrt{\sum_k \|\mathbf{x}_{koop}(kT_s) - \mathbf{x}_{real}(kT_s)\|_2^2}}{\sqrt{\sum_k \|\mathbf{x}_{real}(kT_s)\|_2^2}}, \quad (21)$$

where $\mathbf{x}_{koop}(kT_s)$ and $\mathbf{x}_{real}(kT_s)$ are the vectors of states obtained at the time $t = kT_s$ from the Koopman operator based approximation and the real nonlinear system, respectively. The RMSE was evaluated over 5000 trajectories with the same distribution of initial conditions as the learning dataset, with $T_s = 0.01s$ and each trajectory 30 samples long. For more details on the subject, please refer to [11].

Finally, the polynomial basis function of order $k = 4$ was chosen, with the average $RMSE = 9.7089\%$ and base cardinality number $N_\psi = 35$. A simulation example is shown in Fig.2. In this example, the prediction is restarted every $\Delta t = 0.5s$ to simulate the MPC execution. It can be seen that the Koopman operator results in a better approximation than both the linear system approximated by the first order derivative evaluated iteratively and the one evaluated at $t = 0s$.

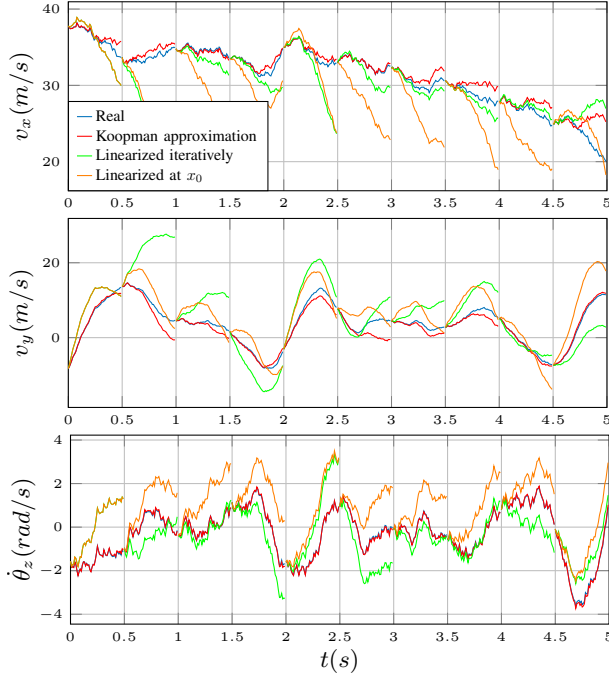


Fig. 2. This figure shows a comparison between the real nonlinear system, the linear system approximated by the Koopman operator with the polynomial basis of order 4, the linear system approximated by a standard first order derivative linearization evaluated iteratively every time when the prediction is restarted (every 0.5s) and a standard first order derivative linearization evaluated at $t = 0s$. The initial state was randomly chosen as $\mathbf{x}_0 = [37.7777 \ -8.2569 \ -1.5693]^T$ and the input signal is randomly sampled from uniform distribution in the interval $[-50000, 50000]$.

IV. MODEL PREDICTIVE CONTROL

MPC is a control strategy that considers both the dynamical model of a system and its constraints. In the MPC, the control action is obtained by solving a constrained finite horizon optimal control problem for the current state of the system at each sampling time. The optimal control input sequence is calculated for a predicted evolution of the system model over a finite horizon, but only the first element of the control sequence is applied. The state of the system is then measured/estimated again at the next sampling time.

In [6] the authors noted that the linear MPC traditionally solves a convex quadratic problem, which allows fast computation and ensures global optimality. On the other hand, the nonlinear MPC solves non-convex optimization problem with multiple local optima and requires much more computational power. The Koopman operator theory gives us a framework to transform complex nonlinear problems into their linear approximations, while being able to predict some of the nonlinear effects better than with standard linearization approaches.

The first step in formulating the KMPC system is to identify the linearized Koopman model as shown in III. Then, we use the obtained approximation to lift the measured system state into a newly formed higher dimensional space and use this linear model for state propagation within the MPC algorithm. After the optimization problem is solved, an input mapping is performed to transform wheel forces into steering angle and

tire slips, which are then applied to the vehicle. The diagram of the described procedure is shown in Fig. 3.

A. Vehicle dynamics Koopman MPC

Formulation of the MPC problem is given by:

$$\begin{aligned} & \underset{\mathbf{U}_t}{\text{minimize}} && J(\mathbf{z}_t, \mathbf{Y}_t^{ref}, \mathbf{U}_t) \\ & \text{subject to} && \mathbf{z}_{k+1} = \mathbf{A}\mathbf{z}_k + \mathbf{B}\mathbf{u}_k, \\ & && \hat{\mathbf{x}}_k = \mathbf{C}\mathbf{z}_k, \\ & && \hat{\mathbf{y}}_k = \mathbf{H}\hat{\mathbf{x}}_k, \\ & && \mathbf{E}_k\mathbf{u}_k \leq 0, \quad k = t, \dots, t+N-1, \\ & && \mathbf{z}_t = \Psi(\mathbf{x}(t)), \end{aligned} \quad (22)$$

with $\mathbf{U}_t = [\mathbf{u}_t, \mathbf{u}_{t+1}, \dots, \mathbf{u}_{t+N-1}]$ representing the input sequence, $\mathbf{Y}_t^{ref} = [\mathbf{y}_t^{ref}, \mathbf{y}_{t+1}^{ref}, \dots, \mathbf{y}_{t+N}^{ref}]$ the reference sequence and $\hat{\mathbf{y}}_k = [v_x \ \dot{\theta}_z]^T$ the output vector. Cost function equals

$$\begin{aligned} J(\mathbf{z}_t, \mathbf{Y}_t^{ref}, \mathbf{U}_t) = & \sum_{k=t}^{t+N-1} \|\mathbf{u}_k\|_S^2 + \|\Delta\mathbf{u}_k\|_R^2 + \\ & \sum_{k=t}^{t+N} \|\mathbf{H}\hat{\mathbf{x}}_k - \mathbf{y}_k^{ref}\|_Q^2, \end{aligned} \quad (23)$$

where $\mathbf{Q} = \mathbf{Q}^T \succeq 0$, $\mathbf{R} = \mathbf{R}^T \succeq 0$ and $\mathbf{S} = \mathbf{S}^T \succeq 0$ are the weight matrices.

The parameters of the controller are:

- *sample time*: $T = 0.01s$
- *prediction horizon*: $N = 10$
- *input constraint*:

$$\begin{aligned} \mathbf{u}_{max} = -\mathbf{u}_{min} &= [F_{fx}^c \ F_{fy}^c \ F_{rx}^c]^T, \\ F_{fx}^c = F_{rx}^c &= \frac{3.5}{100}C_x, \quad F_{fy}^c = \frac{5^\circ\pi}{180^\circ}C_y \end{aligned}$$

- *input rate constraints*:

$$\begin{aligned} \Delta\mathbf{u}_{max} = -\Delta\mathbf{u}_{min} &= [\Delta F_{fx}^c \ \Delta F_{fy}^c \ \Delta F_{rx}^c]^T, \\ \Delta F_{fx}^c = \Delta F_{rx}^c &= \frac{0.75}{100}C_x, \quad \Delta F_{fy}^c = \frac{0.5^\circ\pi}{180^\circ}C_y \end{aligned}$$

- *input cost*:

$$\mathbf{S} = \text{diag} \left(\frac{1}{F_{fx}^c{}^2}, \frac{1}{F_{fy}^c{}^2}, \frac{1}{F_{rx}^c{}^2} \right)$$

- *input rate cost*:

$$\mathbf{R} = \text{diag} \left(\frac{1}{\Delta F_{fx}^c{}^2}, \frac{1}{\Delta F_{fy}^c{}^2}, \frac{1}{\Delta F_{rx}^c{}^2} \right)$$

- *tracking error cost*:

$$\mathbf{Q} = \text{diag} (10^5, 10^5)$$

Input constraints and input rate constraints are determined using corresponding tire slip and slip angle constraints, which is valid for small steering angles.

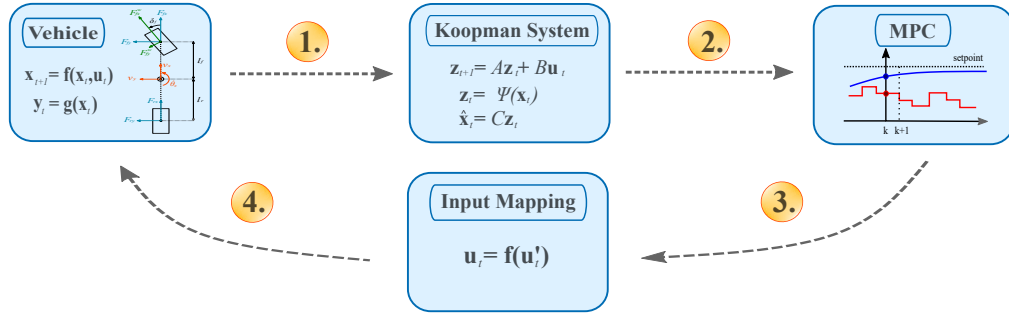


Fig. 3. Diagram showing KMPC framework. The procedure is divided into four separate steps: 1. measure current state $\mathbf{x}_t$, 2. use the identified model to determine the lifted state $\mathbf{z}_t$ and forward it to the MPC, 3. solve the MPC problem and obtain the optimal control input in the form of longitudinal and lateral tire forces, 4. transform the optimal forces into steering angle and tire slip and apply them to the vehicle. All the steps are executed in a loop.

B. Reference generation

The yaw rate reference is estimated using a kinematic bicycle model with zero slip angles on all wheels [16]:

$$\dot{\theta}_z^{ref} = \frac{v_x^{ref}}{l_f + l_r} \tan \delta_f. \quad (24)$$

This model is suitable for describing vehicle motion at low velocities so the controlled vehicle won't be able to achieve the generated references at higher velocities. The reference vector contains longitudinal velocity and yaw rate: $\mathbf{y}^{ref} = [v_x^{ref} \dot{\theta}_z^{ref}]^T$.

C. Input mapping

The tire forces $\mathbf{u}'_t = [F_{fx} \ F_{fy} \ F_{rx}]^T$ obtained from the MPC algorithm have to be transformed into steering angle and tire slips $\mathbf{u}_t = [\delta_f \ s_{fx} \ s_{rx}]^T$, before they are applied to the system. To do this, first (8) - (10) are used to define an equation which describes front tire slip angle $\alpha_f = \alpha_f(\delta_f)$, based on the measured state vector. After that, (5) was inverted and combined with (11) to obtain a set of nonlinear equations:

$$\begin{aligned} F_{fx} &= C_x s_{fx} \cos \delta_f + C_y \alpha_f(\delta_f) \sin \delta_f, \\ F_{fy} &= C_x s_{fx} \sin \delta_f - C_y \alpha_f(\delta_f) \cos \delta_f, \\ F_{rx} &= C_x s_{rx}, \end{aligned} \quad (25)$$

which were then solved using trust-region algorithm [20]. The previously applied input vector $\mathbf{u}_{t-1}$ was used as the starting point.

D. Simulation results

The simulation was run in MATLAB and the MPC optimization problem was formulated with the YALMIP toolbox [21] and solved with the Gurobi solver [22]. The performed manoeuvre is the sine with dwell with a variable longitudinal velocity reference.

Fig. 4 shows that the MPC is capable of controlling longitudinal velocity and yaw rate response of the vehicle, with increased yaw rate tracking error from time $t_1 = 4s$ to $t_2 = 5s$. The optimal tire forces can be seen in Fig. 5, along with the steering angle and the tire slips. Since the tire forces are limited using hard constraints, they are always within the given limits. In addition, the tire slips are also within their

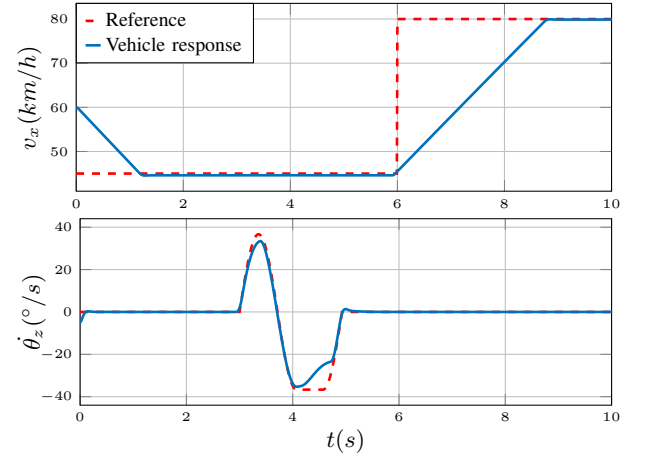


Fig. 4. Longitudinal velocity v_x and yaw rate $\dot{\theta}_z$ response.

limits. This does not need to hold in a general case, but it is true because the small steering angle assumption is fulfilled, as specified in IV-A. This example proves that the proposed approach can be applied for controlling vehicle dynamics.

V. CONCLUSION

In this paper, a Koopman operator based identification of vehicle dynamics using the EDMD was presented. Three state single-track model without a tire model was used to create the dataset, which was then used by the EDMD algorithm to approximate the linear model in a higher dimensional state space. This model showed promising results, as well as the ability to predict the system trajectory better than the models created using Taylor expansion based linearization. An additional verification was performed by including this model in the MPC design and applying it to the original vehicle model. The test proved the efficiency of the proposed method and demonstrated its potential for vehicle motion control.

Possible further research directions include the application of the proposed method to a more complex vehicle model and the development of a more sophisticated approximation method that would be able to account for input nonlinearities. The latter could be achieved e.g. by combining the Koopman operator and/or the EDMD with deep learning.

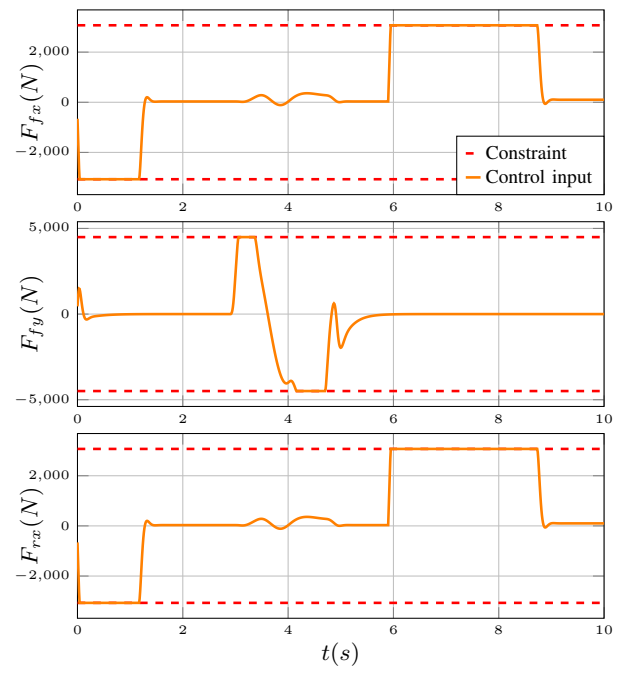
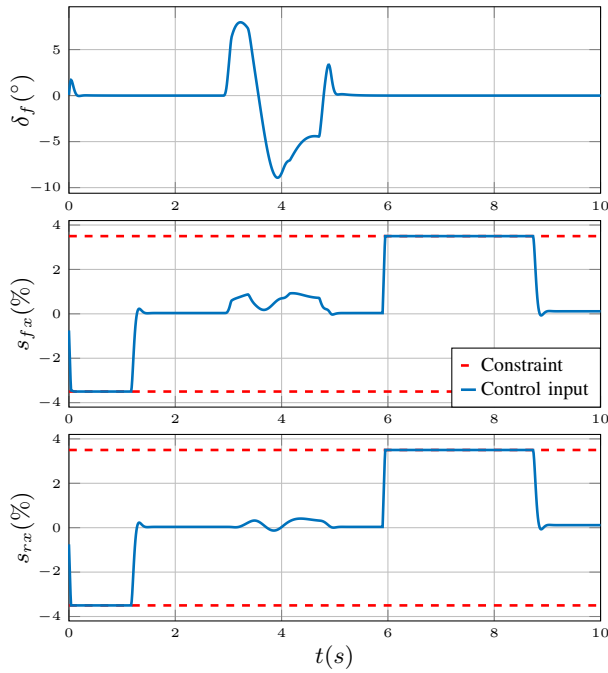


Fig. 5. The figure shows the optimal forces obtained from the MPC (right) together with their hard constraints and final control inputs (left) calculated from forces using input mapping described in IV-C.

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